Identifying Extreme Emission Line Galaxies from VLT MUSE Observations

Understanding the Epoch of Reionization through nearby spectral proxies using integral field spectroscopy of galaxies

T G Hardy

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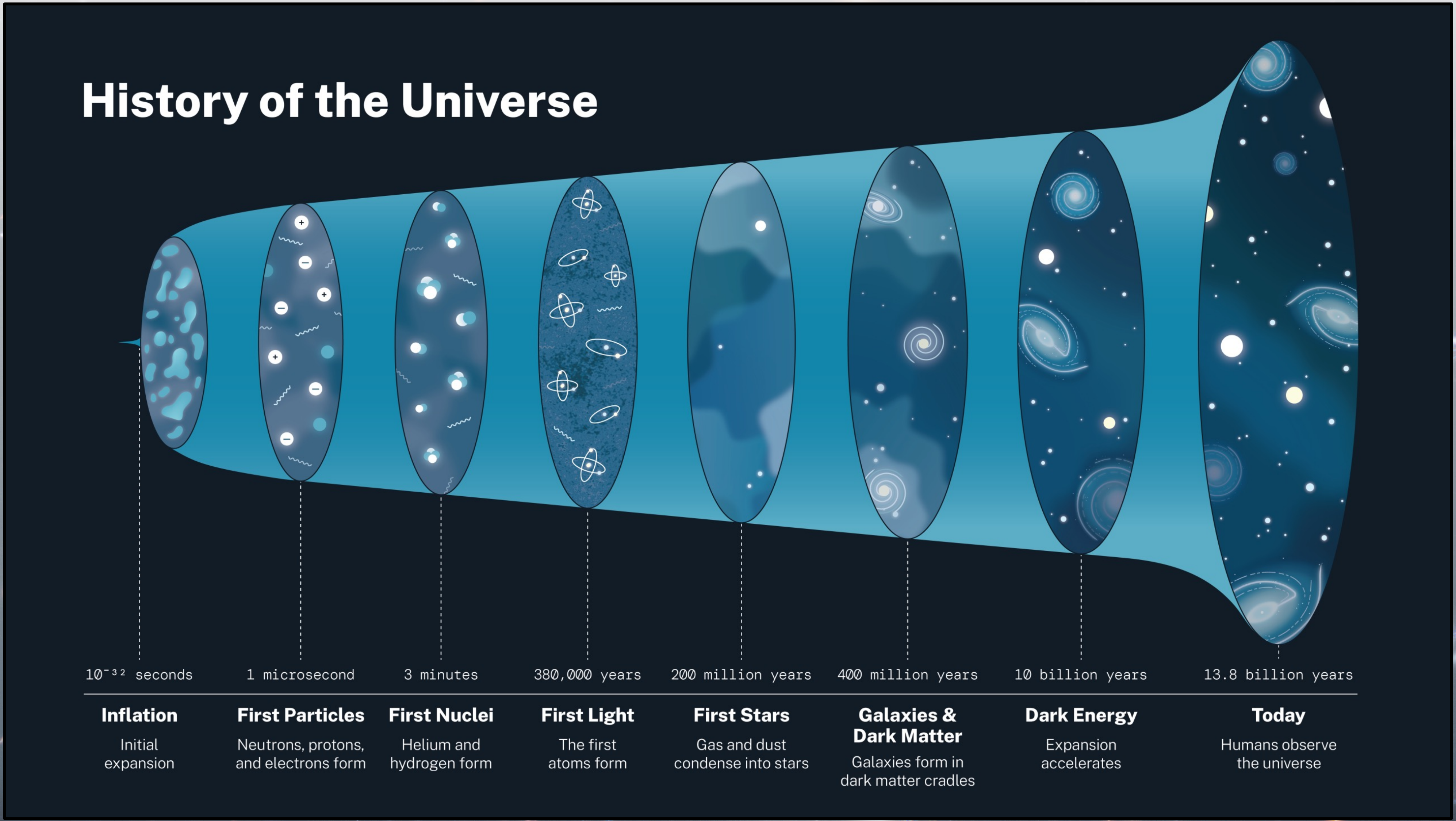
Supervised by Prof Alastair Edge

CEA@ Durham University


History of the Universe
10^{-32} seconds
1 microsecond
3 minutes
380,000 years
200 million years
400 million years
10 billion years
13.8 billion years
Inflation
Initial expansion
First Particles
Neutrons, protons, and electrons form
First Nuclei
Helium and hydrogen form
First Light
The first atoms form
First Stars
Gas and dust condense into stars
Galaxies & Dark Matter
Galaxies form in dark matter cradles
Dark Energy
Expansion accelerates
Today
Humans observe the universe

Spectroscopic confirmation of two luminous galaxies at a redshift of 14

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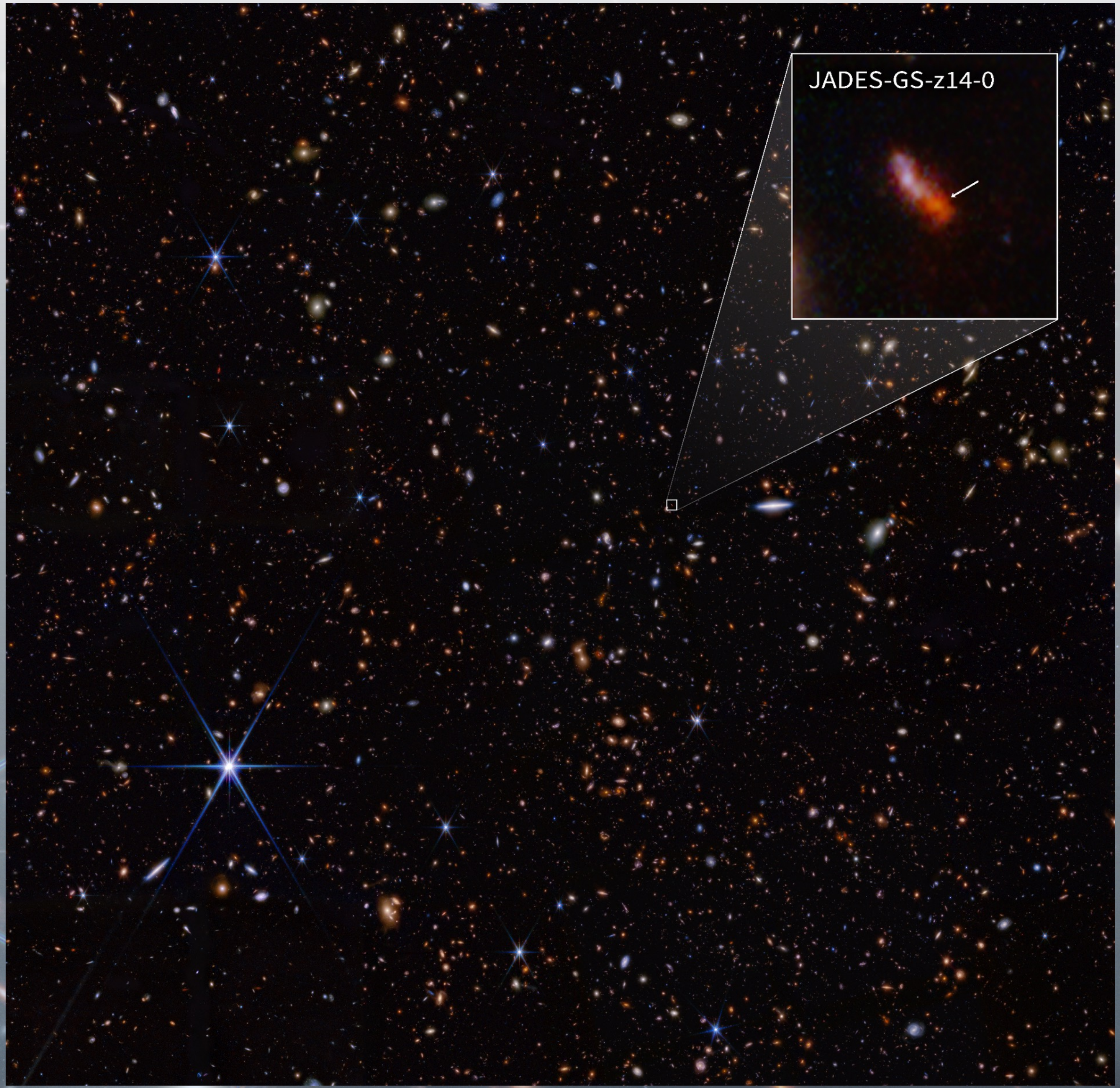
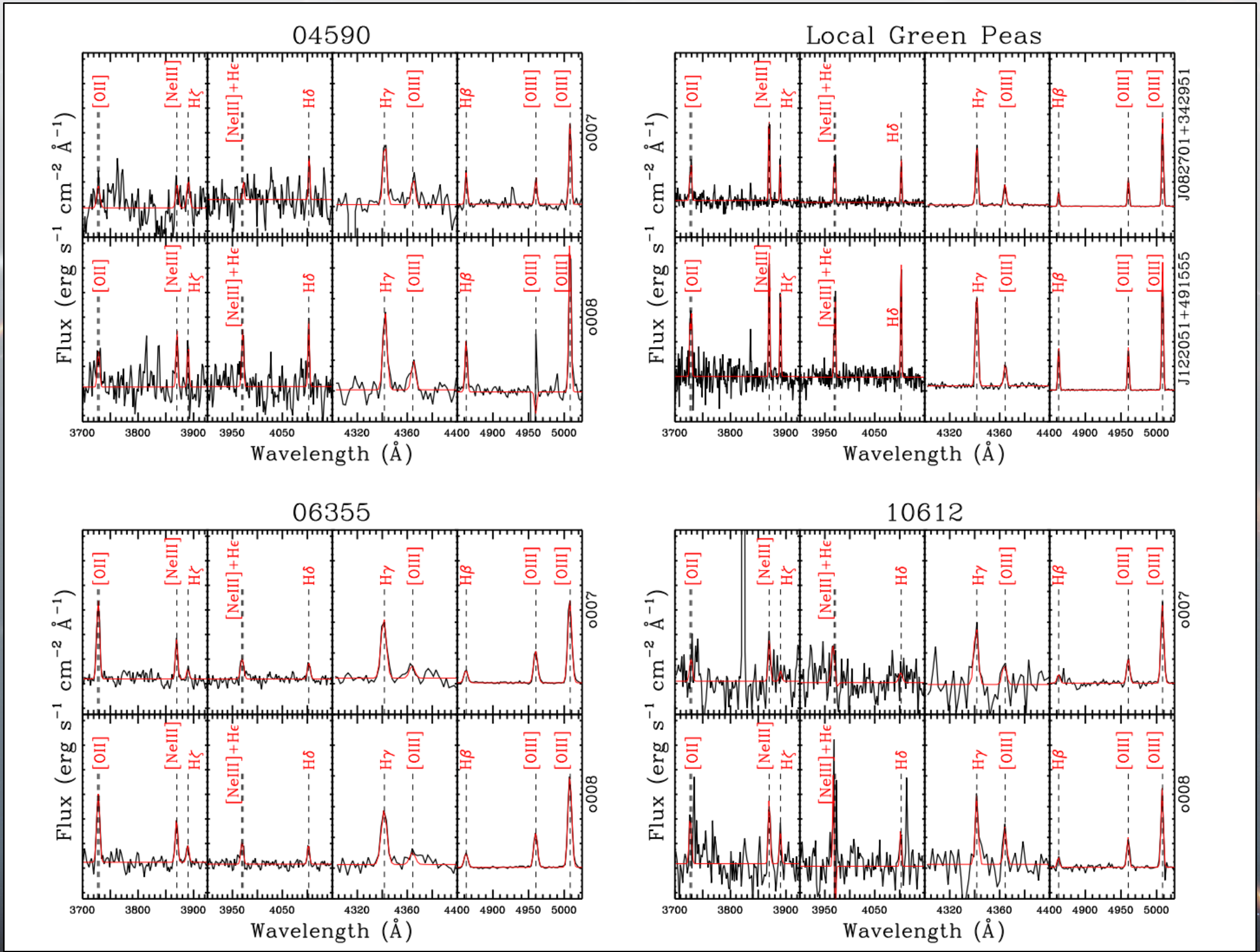
Stefano Carniani[1✉], Kevin Hainline[2], Francesco D'Eugenio[3,4], Daniel J. Eisenstein[5], Peter Jakobsen[6,7], Joris Witstok[3,4], Benjamin D. Johnson[5], Jacopo Chevallard[8], Roberto Maiolino[3,4,9], Jakob M. Helton[2], Chris Willott[10], Brant Robertson[11], Stacey Alberts[2], Santiago Arribas[12], William M. Baker[3,4], Rachana Bhatawdekar[13], Kristan Boyett[14,15], Andrew J. Bunker[8], Alex J. Cameron[8], Phillip A. Cargile[5], Stéphane Charlot[16], Mirko Curti[17], Emma Curtis-Lake[18], Eiichi Egami[2], Giovanna Giardino[19], Kate Isaak[20], Zhiyuan Ji[2], Gareth C. Jones[8], Nimisha Kumari[21], Michael V. Maseda[22], Eleonora Parlanti[1], Pablo G. Pérez-González[12], Tim Rawle[13], George Rieke[2], Marcia Rieke[2], Bruno Rodríguez Del Pino[12], Aayush Saxena[8,9], Jan Scholtz[3,4], Renske Smit[23], Fengwu Sun[2,5], Sandro Tacchella[3,4], Hannah Übler[3,4], Giacomo Venturi[1], Christina C. Williams[24] & Christopher N. A. Willmer[2]

The first observations of the James Webb Space Telescope (JWST) have revolutionized our understanding of the Universe by identifying galaxies at redshift $z \approx 13$ (refs. 1–3). In addition, the discovery of many luminous galaxies at Cosmic Dawn ($z > 10$) has suggested that galaxies developed rapidly, in apparent tension with many standard models[4–8]. However, most of these galaxies lack spectroscopic confirmation, so their distances and properties are uncertain. Here we present JWST Advanced Deep Extragalactic Survey–Near-Infrared Spectrograph spectroscopic confirmation of two luminous galaxies at $z = 14.32^{+0.08}_{-0.20}$ and $z = 13.90 \pm 0.17$. The spectra reveal ultraviolet continua with prominent Lyman-α breaks but no detected emission lines. This discovery proves that luminous galaxies were already in place 300 million years after the Big Bang and are more common than what was expected before JWST. The most distant of the two galaxies is unexpectedly luminous and is spatially resolved with a radius of 260 parsecs. Considering also the very steep ultraviolet slope of the second galaxy, we conclude that both are dominated by stellar continuum emission, showing that the excess of luminous galaxies in the early Universe cannot be entirely explained by accretion onto black holes. Galaxy formation models will need to address the existence of such large and luminous galaxies so early in cosmic history.

Spectroscopic observations from the James Webb Space Telescope (JWST)–Near-Infrared Spectrograph (NIRSpec)[9] have been recently carried out for three candidate galaxies at redshift $z > 14$, selected within the JWST Advanced Deep Extragalactic Survey (JADES) campaigns[10,11]. These galaxies were photometrically identified from within the 58 arcmin² observations of the GOODS-S field through JWST observations with up to 13 NIRCam and seven Mid-Infrared Instrument (MIRI) filters[7,12,13]. On the basis of photometry from the Hubble Space Telescope and Cycle 1 JWST–NIRCam data, the probability of these galaxies being low-redshift interlopers was less than 1% (ref. 12). By happenstance, the brightest of these three candidate galaxies (hereafter JADES-GS-z14-0) is located at a projected distance of only 0.4 arcsec from a foreground galaxy, and this interloper is at a redshift in which its Balmer break is spectrally coincident with the observed photometric Lyman-α break of the distant galaxy. For this reason, and because of its high inferred luminosity at the photometric redshift, JADES-GS-z14-0 was previously considered a low-redshift interloper with a peculiar spectral energy distribution (SED)[12,13]. The 'low-redshift solution' was later disfavoured from the analysis[7] of the JWST–NIRCam observations carried out in the JADES Origins Field (JOF) programme[11], which included additional deep medium-band NIRCam observations that substantially strengthened the case for the source being at high redshift.

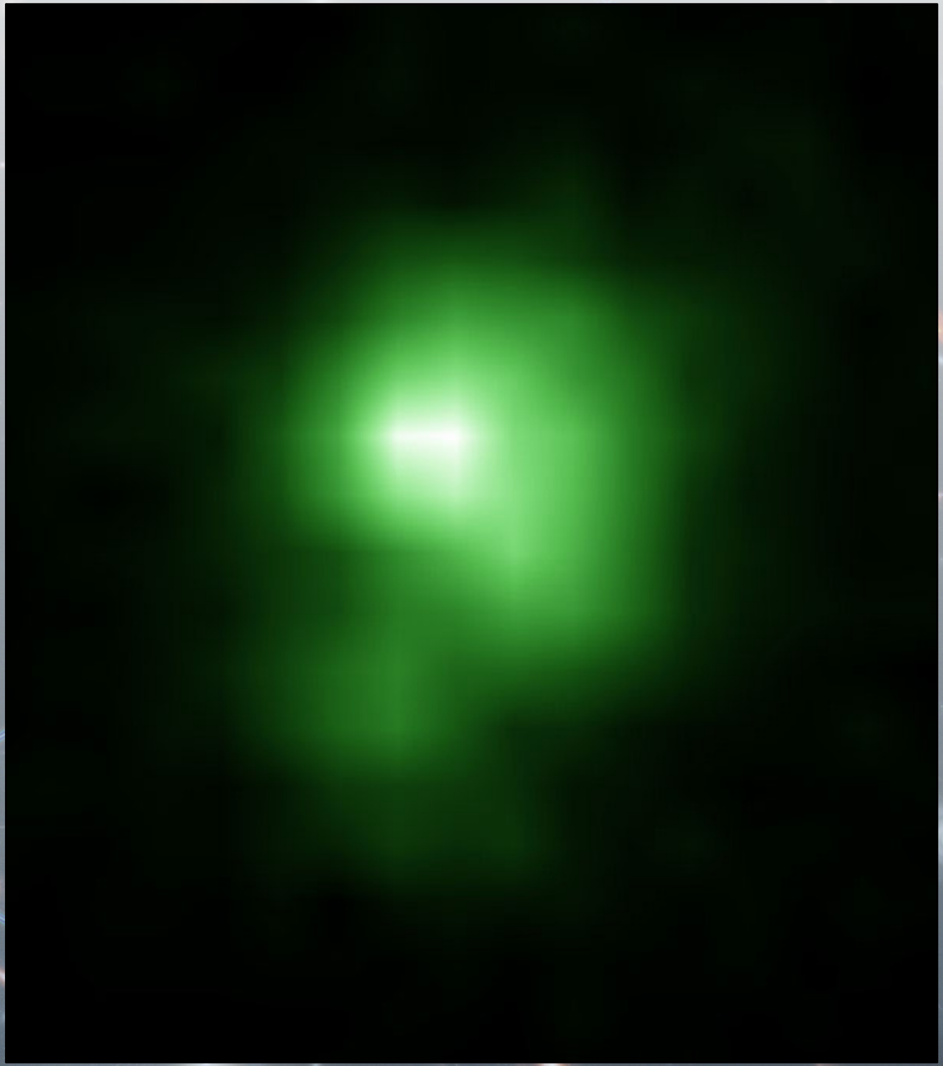
[1]Scuola Normale Superiore, Pisa, Italy. [2]Steward Observatory, University of Arizona, Tucson, AZ, USA. [3]Kavli Institute for Cosmology, University of Cambridge, Cambridge, UK. [4]Cavendish Laboratory, University of Cambridge, Cambridge, UK. [5]Center for Astrophysics, Harvard & Smithsonian, Cambridge, MA, USA. [6]Cosmic Dawn Center (DAWN), Copenhagen, Denmark. [7]Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark. [8]Department of Physics, University of Oxford, Oxford, UK. [9]Department of Physics and Astronomy, University College London, London, UK. [10]NRC Herzberg, Victoria, British Columbia, Canada. [11]Department of Astronomy and Astrophysics, University of California, Santa Cruz, Santa Cruz, CA, USA. [12]Centro de Astrobiología (CAB), CSIC–INTA, Madrid, Spain. [13]European Space Agency (ESA), European Space Astronomy Centre (ESAC), Madrid, Spain. [14]School of Physics, University of Melbourne, Parkville, Victoria, Australia. [15]ARC Centre of Excellence for All Sky Astrophysics in 3 Dimensions (ASTRO 3D), Melbourne, Victoria, Australia. [16]Institut d'Astrophysique de Paris, Sorbonne Université, CNRS, Paris, France. [17]European Southern Observatory, Garching, Germany. [18]Centre for Astrophysics Research, Department of Physics, Astronomy and Mathematics, University of Hertfordshire, Hatfield, UK. [19]ATG Europe for the European Space Agency, ESTEC, Noordwijk, the Netherlands. [20]European Space Agency, ESTEC, Noordwijk, the Netherlands. [21]AURA for European Space Agency, Space Telescope Science Institute, Baltimore, MD, USA. [22]Department of Astronomy, University of Wisconsin-Madison, Madison, WI, USA. [23]Astrophysics Research Institute, Liverpool John Moores University, Liverpool, UK. [24]NSF's National Optical-Infrared Astronomy Research Laboratory, Tucson, AZ, USA. ✉e-mail: stefano.carniani@sns.it




04590
Local Green Peas
06355
10612
Flux (erg s^{-1} cm^{-2} Å^{-1})
Wavelength (Å)
[OII]
[NeIII]
Hζ
[NeIII]+Hε
Hδ
Hγ
[OIII]
Hβ
o007
o008
J082701+342951
J122051+491555
3700
3800
3900
3950
4050
4320
4360
4400
4900
4950
5000

What Are Peas?

- Nearby **highly star forming**
- Spectra dominated by **line emission**
- Very low metallicity
  - PopIII?
  - Young galaxies
- Large **equivalent line widths**
- Strong **Halpha** and **OIII** lines (ionized)



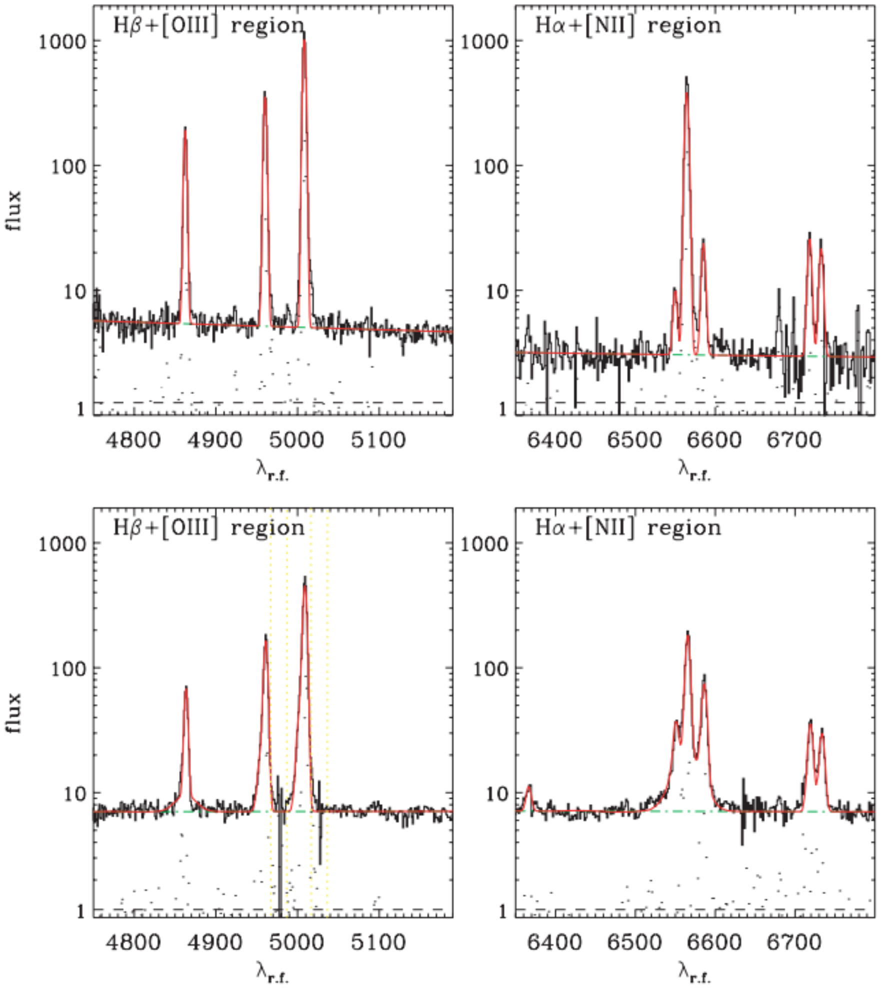


Figure 5. Example spectral fits from GANDALF in the regions around Hα (right) and Hβ (left). The top plots show a typical star-forming Pea and the bottom plots show a typical NLS1.

Figure 1. Example g, r, i composite colour $50 \times 50\,\mathrm{arcsec}^2$ SDSS images classified by Pea hunters, with the r band representing green light. The distinctly green colour and compact morphology makes the Peas (left-hand three images) easily distinguishable from the classical elliptical (right-hand image). The elliptical galaxy is clearly red and has a smooth profile, while the Peas are r band dominated and unresolved in these images, appearing like stellar point sources. All objects shown here are at $z \sim 0.2$.

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The core of the massive cluster merger MACS J0417.5-1154 as seen by VLT/MUSE

Mathilde Jauzac,[1,2,3★] Guillaume Mahler,[4] Alastair C. Edge,[1] Keren Sharon,[4] Steven Gillman,[1] Harald Ebeling,[5] David Harvey,[6] Johan Richard,[7] Stephen L. Hamer,[8] Michele Fumagalli,[1,2] A. Mark Swinbank,[1] Jean-Paul Kneib,[5] Richard Massey,[1,2] Philippe Salomé[9]

[1]Centre for Extragalactic Astronomy, Department of Physics, Durham University, Durham DH1 3LE, UK
[2]Institute for Computational Cosmology, Durham University, South Road, Durham DH1 3LE, UK
[3]Astrophysics and Cosmology Research Unit, School of Mathematical Sciences, University of KwaZulu-Natal, Durban 4041, South Africa
[4]Department of Astronomy, University of Michigan, 1085 South University Ave, Ann Arbor, MI 48109, USA
[5]Institute for Astronomy, University of Hawaii, 2680 Woodlawn Drive, Honolulu, Hawaii 96822, USA
[6]Laboratoire d'Astrophysique, Ecole Polytechnique Fédérale de Lausanne (EPFL), Observatoire de Sauverny, CH-1290 Versoix, Switzerland
[7]CRAL, Observatoire de Lyon, Université Lyon 1, 9 Avenue Ch. André, 69561 Saint Genis Laval Cedex, France
[8]Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB1 0HA, UK
[9]LERMA, Observatoire de Paris, CNRS, PSL University, Sorbonne University, 75014 Paris, France

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ABSTRACT

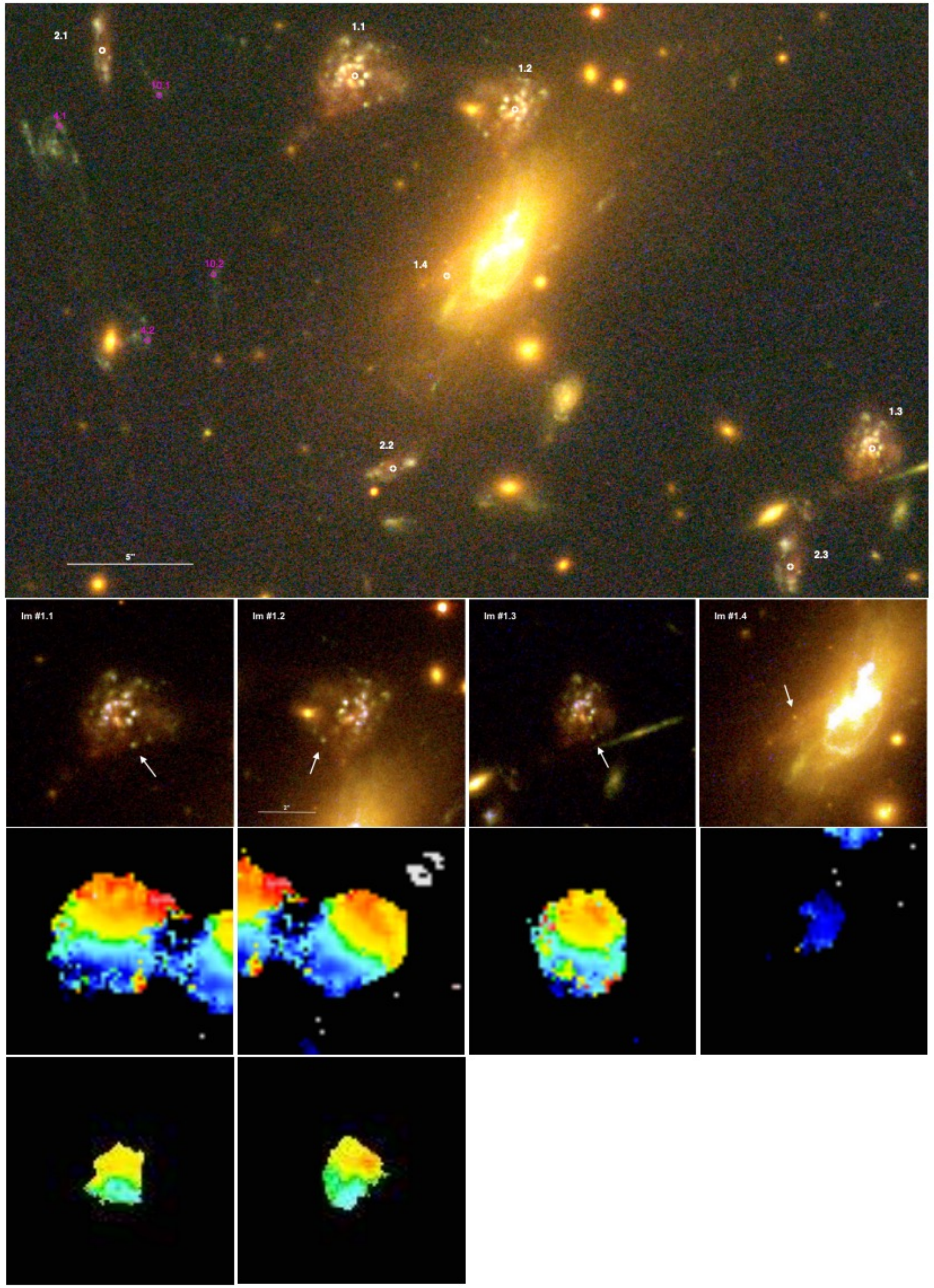
We present a multi-wavelength analysis of the core of the massive galaxy cluster MACS J0417.5-1154 ($z = 0.441$). Our analysis takes advantage of VLT/MUSE observations which allow the spectroscopic confirmation of three strongly-lensed systems. System #1, nick-named *The Doughnut*, consists of three images of a complex ring galaxy at $z = 0.8718$ and a fourth, partial and radial image close to the Brightest Cluster Galaxy (BCG) only discernible thanks to its strong [OII] line emission. The best-fit mass model (rms of 0.38″) yields a two-dimensional enclosed mass of $M(\mathrm{R} < 200\,\mathrm{kpc}) = (1.77 \pm 0.03) \times 10^{14}\,\mathrm{M}_{\odot}$ and almost perfect alignment between the peaks of the BCG light and the dark matter of $(0.5 \pm 0.5)''$. We observe a significant misalignment when system #1 radial image is omitted. The result serves as an important caveat for studies of BCG-dark matter offsets in galaxy clusters. Using *Chandra* to map the intra-cluster gas, we observe an offset between gas and dark-matter of $(1.7 \pm 0.5)''$, and excellent alignment of the X-ray peak with the location of optical emission line associated with the BCG. We interpret all observational evidences in the framework of on-going cluster merger activity, noting specifically that the coincidence between the gas and optical line peaks may be evidence of dense, cold gas cooled directly from the intra-cluster gas. Finally we measure the surface area, σ_{μ}, above a given magnification factor μ, a metric to estimate the lensing power of a lens, $\sigma(\mu > 3) = 0.22\,\mathrm{arcmin}^2$, which confirms MACS J0417 as an efficient gravitational lens.

Key words: Gravitational Lensing – Galaxy Clusters – Individual (MACS J0417.5-1154)

1 INTRODUCTION

Galaxy clusters are the most massive gravitationally bound objects in the observable Universe and are connected to each other on large scales via a web of filaments of matter, commonly referred to as the cosmic web (e.g. Bond et al. 1996). The location of galaxy clusters in this web makes them undergo constant growth through mergers with smaller clusters and groups of galaxies and surrounding material (Springel et al. 2005; Schaye et al. 2015; Colless et al. 2001). This non-equilibrium state allows us to witness evolution and formation processes (Dietrich et al. 2012; Jauzac et al. 2012, 2016b, 2018b) that are not readily observable in other laboratories. Among the many aspects of modern astrophysics and cosmology addressed by cluster studies are the nature of dark matter and its physical properties, galaxy evolution in dense environments, and the influ-

★ E-mail: mn@ras.org.uk (KTS)



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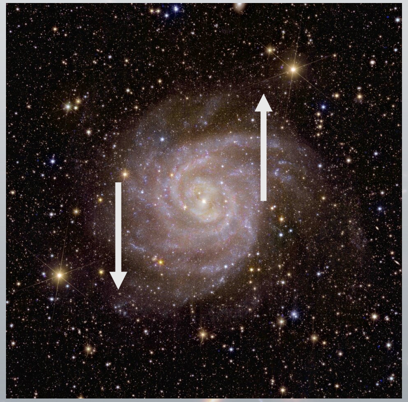
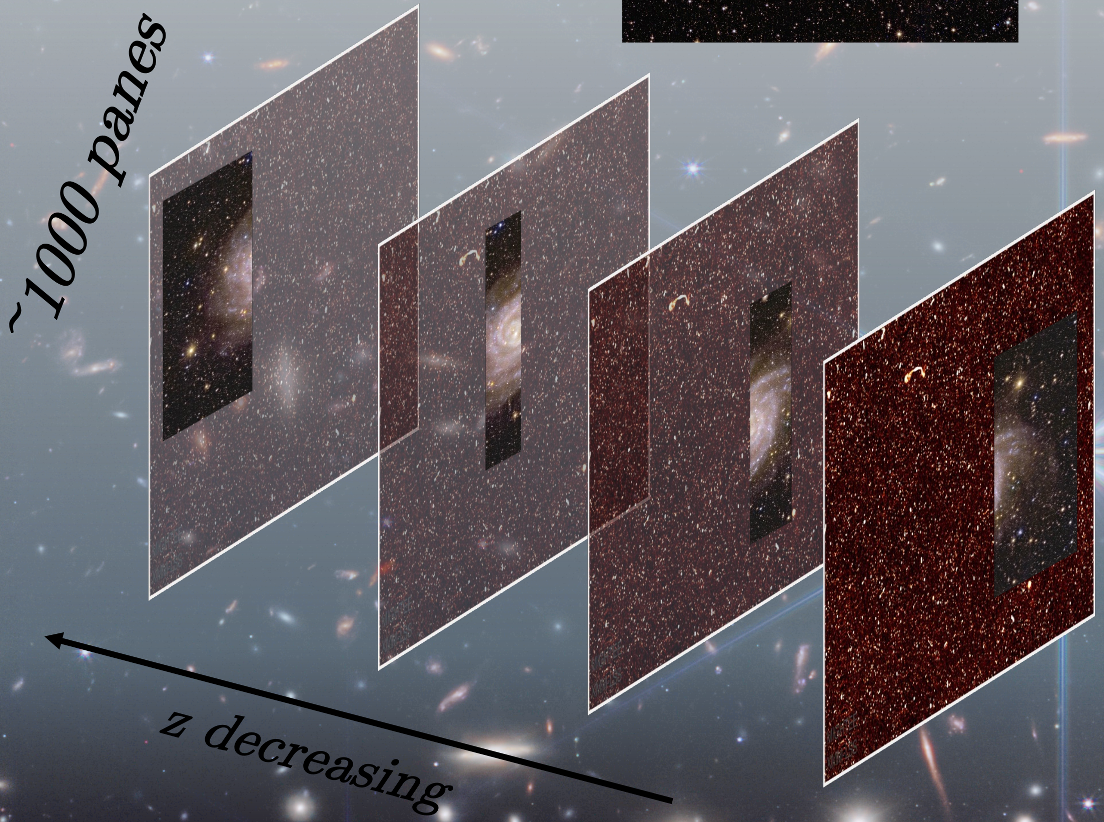


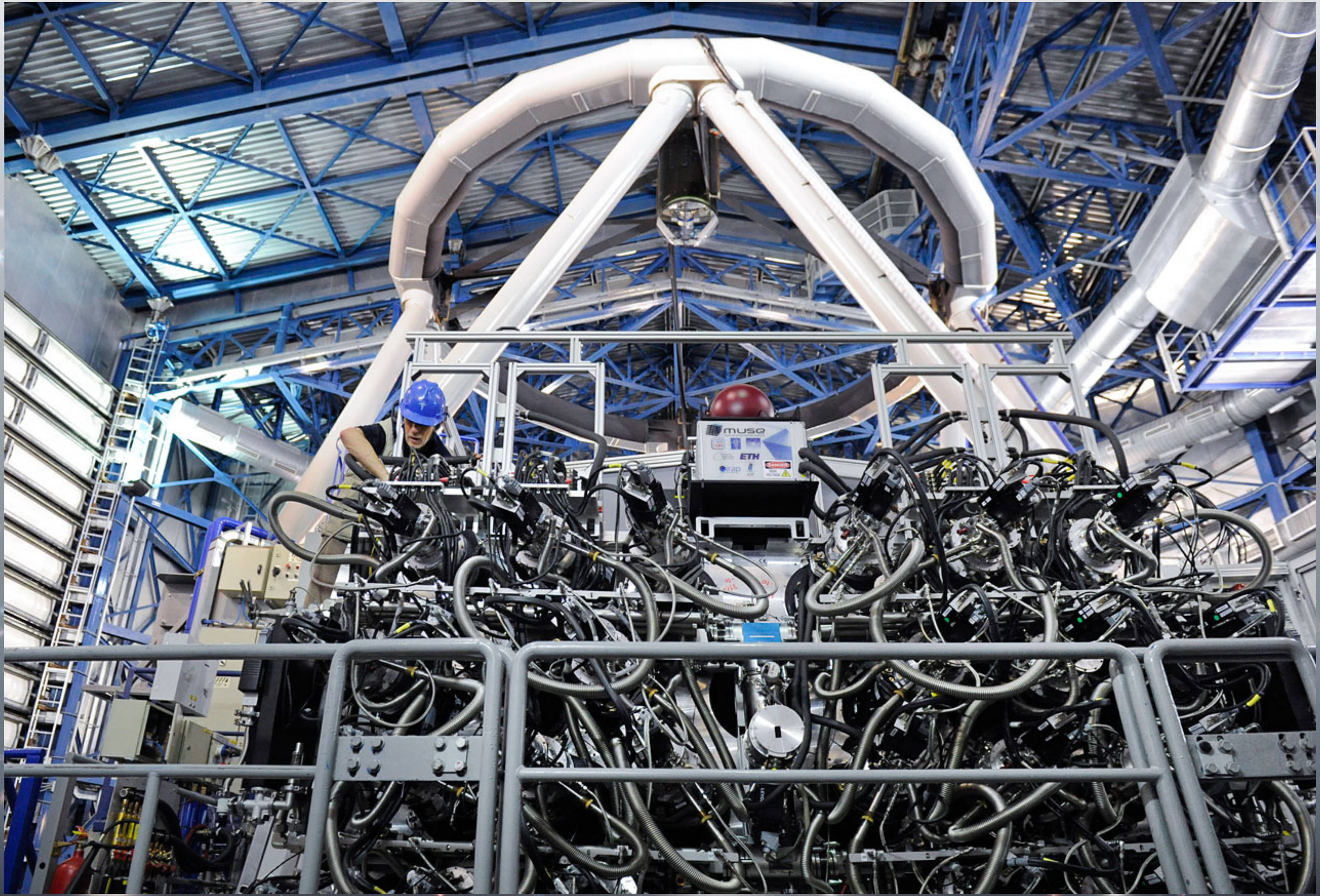


What is a Data Cube?

- ***'bands'*** *of EM frequency*
- *each* ***band*** *has an* ***image***
- *A* ***lot*** *of data; order of* ***9×80GB***

Objects 'move around' in the cube!

8


muse
ETH

How Can We Check?

- Take **spectra** of galaxies using **VLT-MUSE**
- Extract **spectral lines** and calculate **line ratios**
- Use OIII-OII line ratio to then **calculate metallicity**
- (if no OII then use continuum as lower)
- Extract **selection function**

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DIRECT T_E METALLICITY CALIBRATION OF R23 IN STRONG LINE EMITTERS

Tianxing Jiang[1,2,3], Sangeeta Malhotra[4], James E. Rhoads[4], Huan Yang[5]

[1]School of Earth and Space Exploration, Arizona State University, Tempe, AZ 85287, USA; tianxing.jiang@asu.edu
[2]LSSTC Data Science Fellow
[3]Department of Astronomy, University of Maryland, College Park, MD 20742, USA
[4]NASA Goddard Space Flight Center, Greenbelt, MD 20771, USA
[5]Las Campanas Observatory, Carnegie Observatories, La Serena, Chile

ABSTRACT

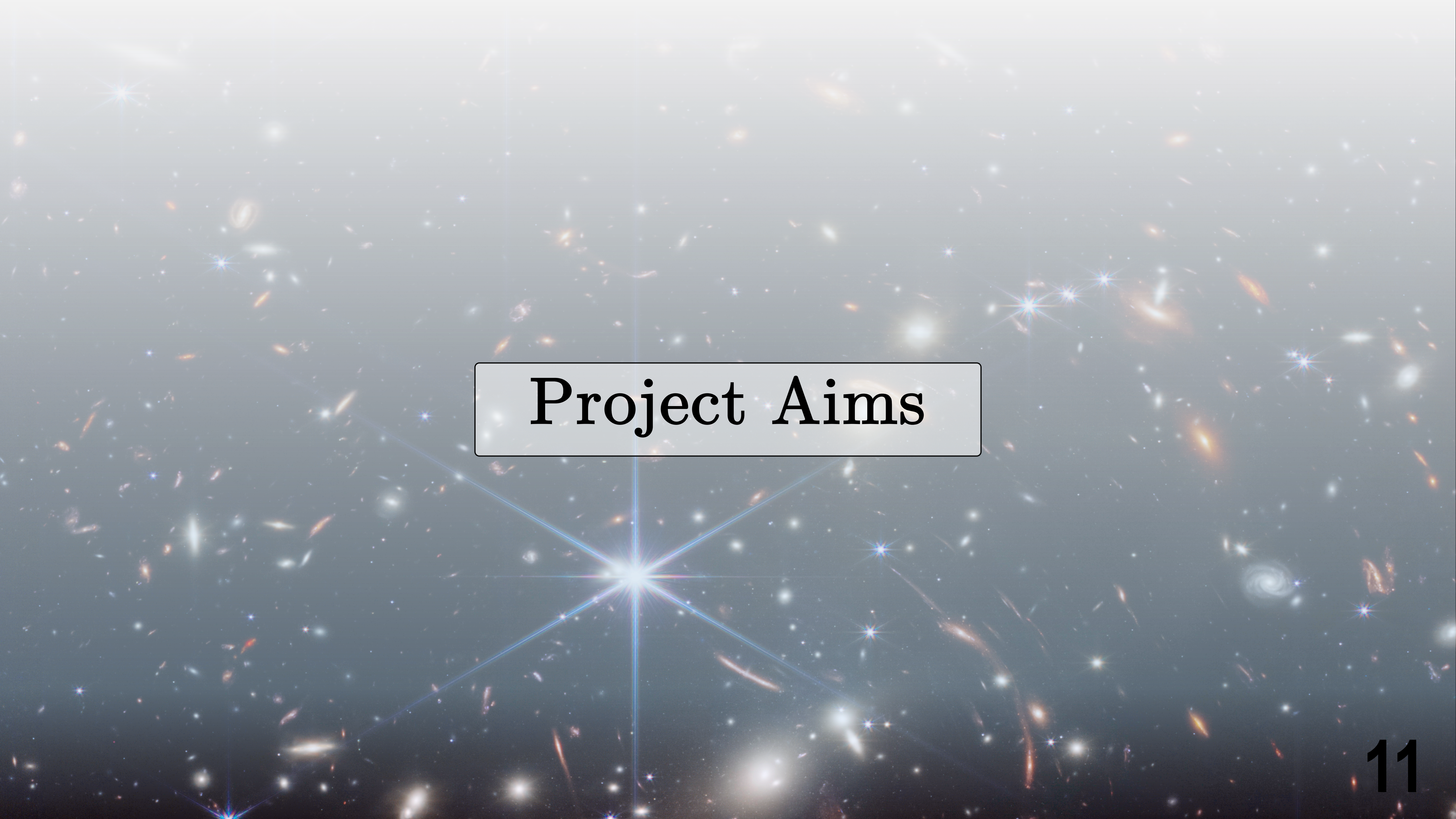
The gas metallicity of galaxies is often estimated using strong emission lines such as the optical lines of [OIII] and [OII]. The most common measure is "R23", defined as ([OII]$\lambda\lambda$3726, 3729 + [OIII]$\lambda\lambda$4959,5007)/Hβ. Most calibrations for these strong-line metallicity indicators are for continuum selected galaxies. We report a new empirical calibration of R23 for extreme emission-line galaxies using a large sample of about 800 star-forming green pea galaxies with reliable T_e-based gas-phase metallicity measurements. This sample is assembled from Sloan Digital Sky Survey (SDSS) Data Release 13 with the equivalent width of the line [OIII]λ5007 $>$ 300 Å or the equivalent width of the line Hβ $>$ 100 Å in the redshift range $0.011 < z < 0.411$. For galaxies with strong emission lines and large ionization parameter (which manifests as log [OIII]$\lambda\lambda$4959,5007/[OII]$\lambda\lambda$3726,3729 $\geq$ 0.6), R23 monotonically increases with log(O/H) and the double-value degeneracy is broken. Our calibration provides metallicity estimates that are accurate to within $\sim$ 0.14 dex in this regime. Many previous R23 calibrations are found to have bias and large scatter for extreme emission-line galaxies. We give formulae and plots to directly convert R23 and [OIII]$\lambda\lambda$4959,5007/[OII]$\lambda\lambda$3726,3729 to log(O/H). Since green peas are best nearby analogs of high-redshift Lyman-α emitting galaxies, the new calibration offers a good way to estimate the metallicities of both extreme emission-line galaxies and high-redshift Lyman-α emitting galaxies. We also report on 15 galaxies with metallicities less than 1/12 solar, with the lowest metallicities being 12+log(O/H) = 7.25 and 7.26.

Keywords: ISM: abundances — galaxies: abundances — galaxies: evolution — galaxies: starburst

1. INTRODUCTION

In the galactic ecosystem, stars form from the collapse of gas clouds and fuse hydrogen and helium into heavy elements (metals); stars eject gas and metals into the interstellar medium by stellar feedback; cool gas in the circumgalactic and intergalactic medium flows into the galaxy; and gas enriched with metals in the galaxy can be transported into the intergalactic medium by galactic outflows. The fraction of gas that has been converted to heavy elements, which is often quantitatively characterized by "metallicity", is key for understanding the star formation history and galactic chemical evolution. In addition, metallicity impacts the luminosity and color of the stellar light, the cooling of gas, and the amount of dust, which in turn determines the interstellar extinction. Robust metallicity measurement is the foundation for investigating mass-metallicity and mass-metallicity-SFR relations and their redshift evolution.

The gas-phase oxygen abundance is usually measured as a good proxy of the metallicity in the interstellar medium of galaxies, since oxygen is the most abundant metal and the emission lines from the most important ionization stages of oxygen can be easily observed in optical. Reliable metallicity measurement of the ionized gas in galaxies requires the measurement of the electron temperature from the ratio of the auroral to the nebular emission lines, such as [OIII]$\lambda\lambda$5007,4959/[OIII]λ4363. However, it is difficult to detect the [OIII]λ4363 line, as it is intrinsically weak. This line is too weak to be observed in metal-rich environments (due to low electron temperature) or faint galaxies. When [OIII]λ4363 lines (or their analogs) are not detected, metallicity-sensitive ratios of strong emission lines are widely used as metallicity indicators (strong-line methods), such as [NII]λ6584/Hα, ([OII]$\lambda\lambda$3726, 3729 + [OIII]$\lambda\lambda$4959,5007)/Hβ (R23), [OIII]λ5007/[NII]λ6584, [SII]$\lambda\lambda$6716, 6731/Hα, [NII]λ6584/[SII]$\lambda\lambda$6716, 6731. Strong-line methods are especially common in studies of high-redshift galaxies (e.g., Erb et al. 2006; Mannucci et al.2010;



Project Aims

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Understanding the Epoch of Reionization
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